Auditable Long-Term Memory: A Deterministic Retrieval
Chain at 95.4±0.4% on LongMemEval-S

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Abstract

We report 479/500 and 475/500 across two independent full passes on LongMemEval-S (Claude Opus reader; CLI alias resolving to `claude-opus-5`, §5.3) under the benchmark's official GPT-4o judge, against a published state of the art of 478/500 (95.60%, Chronos High, PwC, arXiv 2603.16862) — one pass above it, one below: a statistical tie whose spread sits inside the instrument's measured noise band. A second, grok-4.6 reader configuration measured 476/474 on the identical substrate. The system is a deterministic retrieval chain — dense retrieval, cross-encoder reranking, a coverage-first packet compiler, and mechanically extracted reasoning scaffolds, all deterministic code validated by negative controls — with a replaceable LLM reader confined to the final answering step. At margins of a few questions, measurement is the result: the pair's entire spread traced to eight verdict-flip rows; cross-judge agreement is 98%; we observed official-judge verdict flips on byte-identical answers; and a maximum-reasoning-effort reader variant scored *lower* (461/465), with 4 of its 19 per-pass losses being the same answer text judged oppositely. We release every reader output, judge verdict, and control receipt, plus an error dossier that attributes all remaining failures — including two strict gold-answer defects, an improvement stage rejected by our own negative control, and the routes that did not work. A full re-score under the official judge costs about $1.28.

1 Introduction

Long-context memory benchmarks are typically reported as single numbers from single runs, judged by an LLM, with no disclosure of run-to-run or judge variance. At the top of a leaderboard, where entries are separated by a handful of questions out of 500, this practice makes the ranking partially an artifact of measurement noise.

This report makes two claims. **A capability claim**: a deterministic, auditable retrieval chain — in which the language model is confined to a replaceable reader role — measures 479/475 per 500 on LongMemEval-S, pass 1 above the published state of the art, the pair a statistical tie with it, with overlapping confidence intervals and per-run variance comparable to the gap. **A methodology claim**: the apparatus that produced the number is the more important contribution. Every stage below the reader is deterministic code validated by negative controls; the promotion rule was frozen before the deciding run; measurement noise from both the reader and the judge is quantified and disclosed rather than absorbed into the headline.

The system under test is the retrieval core of Archivist, a local-first conversational memory product.

Nothing in the chain is benchmark-specific except the evaluation harness itself.

1.1 Related work

Agentic memory systems (MemGPT, Packer et al., 2023; production systems such as Mem0 and Zep) focus on memory *stores* and management policies; our contribution is orthogonal — determinism below the reader, with the store fixed. Long-term memory benchmarks include LongMemEval (Wu et al., 2024), used here, and LoCoMo (Maharana et al., 2024); we evaluate one benchmark and claim no cross-benchmark generality (§7). The retrieval stack follows the retrieve-and-rerank tradition (Nogueira & Cho, 2019; BGE-M3 family, Chen et al., 2024); reciprocal rank fusion (Cormack et al., 2009) is among the extension-ordering baselines swept in §3.3. On measurement: a self-consistency step (Wang et al., 2023) was built and later withdrawn when internal review showed its two-pass framing was a derived identity (§4.4); the judge-variance study operationalizes known LLM-as-judge concerns (Zheng et al., 2023); the two-pass reporting rule answers the single-run critique of Dodge et al. (2019).

2 Benchmark and evaluation protocol

LongMemEval-S (Wu et al., 2024) contains 500 questions over long multi-session chat histories: 470 answerable questions across six types (single-session-user, single-session-assistant, single-session-preference, multi-session, temporal-reasoning, knowledge-update) and 30 abstention questions whose correct behavior is to decline to answer. Evidence for a question may be spread across sessions recorded weeks apart.

Judge. We use the benchmark's official protocol: GPT-4o (`gpt-4o-2024-08-06`) with the paper's rubric templates. One header sentence of the single-session-preference template was reconstructed from the benchmark's published rubric text; the templates as used are SHA-256 pinned (`9c9d67fab129…`) and released verbatim, so the reconstruction is auditable rather than asserted. Every scoring run executes 12 live positive and 12 live negative control verdicts against a 0.9 pass threshold (preference rows are excluded from the positive control arm because their rubric is the reconstructed one); every run in this report returned 12/12 and 12/12. The harness records the resolved model snapshot per verdict and fails closed on authentication errors. A full 500-row re-score costs $\approx$ \$1.28.

Comparison point (verified at source, 2026-08-31). The published raw-score state of the art is **Chronos High: 478/500 (95.60%)** (PwC, arXiv 2603.16862, Mar 2026; generator Claude Opus 4.6, LongMemEval judge protocol, per-category table published — their Table 2 and Appendix A, Table 4). Two widely-quoted higher percentages are not raw-comparable: Mastra OM's "94.87%" is a category-unweighted average — their own per-category counts sum to **468/500 raw** (generator gpt-5-mini) — and OMEGA's "95.4%" is likewise task-averaged with raw 466/500, judged by GPT-4.1 rather than the canonical judge. One higher claim (481/500, "agentmemory V4") exists in an unreviewed personal repository; we note it without treating it as the published bar. Reader tier varies freely across published entries (GPT-4o through Opus 4.6; our headline reader resolves to Opus 5, §5.3); we follow the field convention of reporting our strongest configuration alongside the full reader ladder.

3 System architecture

The chain has five stages. Stages 1–4 are deterministic: byte-identical outputs across runs are enforced by verification gates, not assumed (verified on the pinned environment; hardware and library versions are recorded in the release).

3.1 Dense retrieval. Per-question dense embedding retrieval over the haystack produces a ranked candidate pool (local inference; pool ceiling: 468/470 questions have all gold sessions somewhere in the pool).

3.2 Cross-encoder reranking. A cross-encoder (`BAAI/bge-reranker-v2-m3`) scores each candidate session against the question; session score is the maximum chunk score (chunk = date header + turn). Scoring is rank-deterministic (fp16, fixed batch, tie-break by prior rank).

3.3 Coverage-first packet compilation. For each question a reading packet is compiled under a hard budget (16 sessions / 40k proxy tokens): the dense baseline top-10 is preserved as a byte-identical prefix, remaining slots fill in cross-encoder order with term-preserving extractive compression, and budget overruns drop the lowest-ranked extension first. Validated by determinism, gold-permutation and question-ID-sabotage negative controls, and budget accounting. Packets are gold-complete for 462/470 questions; a sweep of all five extension-ordering policies confirmed the shipped ordering is globally optimal (462 vs. 454 best alternative).

3.4 Deterministic reasoning scaffolds ("facts"). For 240 of 470 questions the chain emits a deterministic evidence index computed from the packet: session chronology, a canonical user-statement recency domain with clock-granular tie keys, a value-preference invariant, candidate enumeration with quantity roll-up for counting questions, and granularity/unit and event-anchor rules (contract `v3.4-facts-20260831`). Scaffolds are code with in-code coupling invariants that raise on internal inconsistency. The scaffold is gold-blind and question-ID-independent by construction.

3.5 Replaceable reader. The packet, scaffold, and question go to an LLM reader, deliberately replaceable; we measure the chain under multiple readers (§5.3).

4 Measurement methodology

4.1 Two-pass rule (pre-registered). A nondeterministic reader means a single full run at a small margin is not a reportable number. Our promotion rule, adopted before the deciding experiments: a headline requires two full passes agreeing, or one number with a stated ± spread from measured flip variance.

4.2 Judge variance is measured, not assumed. We re-judged both passes of our prior configuration with a second judge on identical rows: agreement 98.2%/97.8%, the second judge strictly harsher. We also observed the official judge returning opposite verdicts on a byte-identical answer across runs. At these margins the judge is the measurement floor; we log such rows rather than resolve them by re-rolling.

4.3 Negative controls at the improvement layer. Every proposed improvement runs against a 60-row control set of baseline-correct questions before integration. This rule has teeth: a verifier stage that recovered 3 previously-wrong rows was *rejected* because the control replay showed it flipped 11 of 59 correct drafts to wrong (§6.2). The reader-tier upgrade in this report passed the same control with zero flips before its full pair was launched.

4.4 Self-consistency tie-break (built, then withdrawn). The raw pair's spread came from eight verdict-flip rows, and a tie-break rule frozen before any additional read gave each flip row three more reads with a five-vote majority. The resulting number was WITHDRAWN from any claim: internal review showed the construction wrote one derived verdict into both passes (an identity, not two agreeing measurements) and its flip-row selection was judge-conditioned. The artifacts remain released as a record; no self-consistency number is claimed anywhere in this report.

4.5 What we did not do. No per-question prompt tuning against gold; no judge re-rolling; no selective reporting (failed routes are in §6); no counting of gold-defect adjudication in any headline number.

5 Results

5.1 Headline

Configuration	Pass 1	Pass 2
v3.4, Claude Opus reader (CLI lane)	**479/500**	**475/500**
v3.4, grok-4.6 high (raw pair)	476/500	474/500
v3.4, grok-4.6 xhigh (agentic transport)	461/500	465/500

The Opus pair is the headline: pass 1 exceeds the published SOTA (478) on the raw score; the pair's both-pass-stable floor is 473 (vs. the grok pair's 471); pass 1 is perfect on three of six answerable types (single-session assistant, user, preference). Its 8 flip rows (6 lost — including 2 abstentions and 2 dossier-listed flip-prone rows — 2 gained) match the flip-noise structure of every other pair here. Under the pre-registered two-pass rule this is a **statistical tie with the published SOTA, one pass above it**, not a clean beat: we state 477±2 and decline the stronger claim our own bar forbids (a pre-release re-judge of pass 1 under the same official judge returned 478 — three verdict flips on identical text, §8 — so even "one above" is a judge roll). The two gold-defect rows from the dossier (§6.3) sit inside this pair as well: one is scored correct via the fabrication the dossier documents (the reader supplies the $500 no session contains), one is lost to fidelity (the evidence-faithful answer contradicts defective gold) — net zero per pass. Under adjudication the pair reads 478/474 (ex-fabrication) to 480/476 (restored); the tie with the published state of the art holds in every stance. Judge controls 24/24 in both passes; a 25-row probe preceded the pair (11/25 stable-wrong recovered, one row no prior reader had ever solved).

The xhigh variant is a measured regression (−15/−9), reported in full. Its gate evidence had looked favorable — a 25-row probe on the high pair's wrong set recovered 9, and a 60-row negative control on baseline-correct rows showed zero flips — but the full pair exposed the gates' limits: only 4 probe recoveries held, and the true correct-row flip rate (∼3.4%) was small enough that a 60-row control had a ∼13% chance of showing zero flips. Row-level attribution of pass 1's 19 losses: 4 are judge flips on answer text identical to the high run's, and 15 share one over-skepticism signature — wrong abstentions on answerable rows, over-filtered counts, hedged golds. More reasoning effort made the reader second-guess correct evidence. (One v1 loss was a staleness artifact, not a reader loss: the reader resumed one row eleven minutes after the judge had scored it on an empty response; a surgical v2 re-judge under fresh 12/12 controls restored it, 460 → 461.)

Prior-configuration detail: answerable 450/470 and 447/470; abstention 26/30 and 27/30. Judge controls passed in every scoring run. Under the gold-defect adjudication scenario (§6.3) the prior pair reads 478/476; we report that as a scenario, never as a score.

5.2 Per-question-type breakdown (official GPT-4o judge)

Question type	n	Raw p1	Raw p2	xhigh p1 / p2	Opus p1 / p2
single-session-assistant	56	56 (100.0%)	56 (100.0%)	55 / 55	56 / 56
single-session-user	64	63 (98.4%)	63 (98.4%)	61 / 60	64 / 64
knowledge-update	72	70 (97.2%)	71 (98.6%)	69 / 70	69 / 70
single-session-preference	30	29 (96.7%)	29 (96.7%)	25 / 25	30 / 28
temporal-reasoning	127	122 (96.1%)	122 (96.1%)	121 / 123	123 / 123
multi-session	121	110 (90.9%)	106 (87.6%)	103 / 106	110 / 109
abstention	30	26 (86.7%)	27 (90.0%)	26 / 26	27 / 25
Total	**500**	**476**	**474**	461 / 465	**479 / 475**

Five of six answerable types are pass-stable to within one question; the raw pair's entire answerable spread is one type (multi-session), partially offset by knowledge-update. Against the published SOTA's category table on its own abstention-folded basis (n: KU 78, MS 133, SSA 56, SSP 30, SSU 70, TR 133 — their Appendix A), our pass-1 gap is −3 knowledge-update and −1 preference, partially offset by +2 multi-session (120/133 vs. their 118/133); pass 2 trails on multi-session as well (116/133), so our multi-session score exceeds theirs in pass 1 only.

5.3 Ablation ladder (single ruler: official GPT-4o judge)

Stage	Score /500
Dense retrieval + flat reading (Flash reader, v2 prompts)	452/499[1]
+ chain, v3.3 operators, Sol reader	468
+ cross-encoder rerank (v3, grok reader)	473
+ v3.4 operators (official pair, grok high)	476 / 474
+ reader xhigh + agentic transport	461 / 465 (regression)
+ reader tier: Claude Opus, same substrate	**479 / 475**

Ex-dossier (all eight gold-defect/boundary rows removed from both readers): grok leads 475/474 vs. Opus 474/471 — Opus's net margin over grok is carried by gold-idiom reading on dossier rows, not by evidence reading on ordinary rows. The Opus reader is the CLI alias `opus` (claude-cli lane, subscription), not a pinned snapshot: no run artifact records the run-time resolution, and a same-day probe (2026-09-01, Claude Code CLI 2.1.258; the runs used 2.1.252) resolved `opus` to `claude-opus-5` — one model generation newer than the Chronos comparator's Opus 4.6. The grok and GLM readers are named models. Three of Opus's abstention credits (`2133c1b5_abs`, `c8090214_abs`, `f685340e_abs`) are premise-repair-then-answer responses the official judge accepts and the GLM cross-judge rejects; under strict abstention the pair reads 476/472 — still a tie. [1]One row of the legacy baseline failed its reader lane and was never re-read; counted as scored-out-of-499 rather than imputed.

5.4 Reader robustness

On identical packets and scaffolds: **Claude Opus 479/475**; grok-4.6-high 476/474; GPT-5.6 (Sol) 455 — measured decomposition versus its prior-contract 468: −5 per-run reader variance on byte-identical non-scaffold rows, −5 negative scaffold×reader interaction, −3 abstention; judge controls 24/24. DeepSeek V4 Pro scores 436/500 (407/470 answerable) with the best abstention behavior of any reader measured (29/30). A 30B floor probe (Nemotron-3.5-Lightning) collapses to 93/500 on the identical gold-complete packets — the chain does not rescue a reader below a capability threshold. We controlled for a serving artifact: correctness barely varies with packet size (28% vs. 15% across size buckets, vs. 90/86% for V4 Pro), and a question-first re-prompt of a failed row was still ignored in favor of distractor-session content — the collapse is model-side, not provider truncation. A gemini-3.7-flash run reached a 189-row partial (a type-ordered prefix, not a sample: single-session-user 64/64, multi-session 73/83, preference 21/30; no other types) and is reported as a bounded signal only. A GLM-5.3 run on the identical substrate scores 471/500 (445/470 answerable, 26/30 abstention; official GPT-4o judge, controls 12/12) — a single-pass curve datapoint per the two-pass rule, not a headline. A Kimi K3 run (the long-context-specialist family) scores 465/500 (444/470 answerable, 21/30 abstention; same judge, controls 12/12) — long-context specialization does not transfer to memory-benchmark dominance on this substrate. Together the fixed substrate exposes a full reader-capability curve (93 → 436 → 455 → 465 → 471 → 476): the packet and scaffold layers carry the score for capable readers, and reader choice moves it by tens of points across tiers but only a few points within a tier. The v3.4 scaffold contract is tuned to the grok reader family (§7).

6 Error analysis: every remaining failure, attributed

Of 470 answerable questions, 24 fail in at least one prior-configuration pass.

6.1 Reader-cognition residue (∼4 rows). The packet contains the correct evidence; the reader commits to an evidence-backed distractor. The residue is question-boundary semantics (does a stated *plan* update a location? does the offered property count as "viewed"?), not evidence selection.

6.2 The verifier our own control rejected. A cite-or-revise verifier recovered 3/12 winnable wrong rows — and the mandatory negative control showed it revised 19/59 baseline-correct drafts, flipping 11 to wrong while repairing none. Blocked by the pre-committed acceptance rule. A second design (draft-blind candidate enumeration + rule-based adjudication) recovered 0/9. Both failures are published; they map the limit of prompting-side repair.

6.3 Gold-answer defects (2 strict, 6 boundary). Documented row-by-row in the released dossier. Strict A: gold requires a $500 expense that appears nowhere in the evidence, plus an event dated after the question date attended in the past tense; a weaker reader that *hallucinated* the missing $500 was scored correct by the official judge. The grok reader's evidence-faithful $220 scores 0 in both passes; we keep the 0. The headline Opus reader, by contrast, supplies the $500 and is scored correct on both passes — the disclosed fabrication point inside 479/475 (§5.1). Strict B: gold says "15 weeks" between two user-dated events 81 days apart. Six further rows are boundary-semantic and claimed as worth nothing. The Chronos authors independently document a defect on question `6d550036` (their §3.5, alongside a judge-variability example on `75f70248`), consistent with our dossier.

6.4 Retrieval ceiling and coverage, measured precisely. One row's gold never reaches the candidate pool; three rows lost an in-pool gold session to packet budget ordering (no global ordering recovers them without losing more). An entity-linking retrieval arm was built, tested (17/17), and retired when measurement showed zero of its added sessions were gold. Remaining failures are measured flip noise.

7 Limitations

- **Margins at this level are not statistical superiority.** Wilson 95% intervals: our 479 [93.7%, 97.2%] and 475 [92.7%, 96.6%] (grok pair: 476 [93.0%, 96.8%], 474 [92.5%, 96.4%]) vs. Chronos's 478 [93.4%, 97.1%] — overlapping almost entirely. (The xhigh pair sits lower: 461 [89.5%, 94.2%], 465 [90.4%, 94.9%].) Nearby entries' intervals overlap substantially;

no comparison entry publishes per-question verdicts, so McNemar is not computable. Our number is reproduced with variance disclosed; comparison entries are single runs.

- **The judge is the floor.** ∼2% verdict variance between judges, and observed flips on identical answers, bound what any entry at these margins can honestly claim.
- **Scaffold–reader coupling.** The v3.4 scaffold is tuned to one reader family; the exact number moves a few points across readers, though the chain's ranking of readers is stable.
- **One benchmark.** No cross-benchmark generality claim.

8 Reproducibility

Released with this report. **Release gate: these claims are verifiable only once the evidence-repository URL resolves here; until then this is a pre-release copy and every "released" statement below is a commitment, not evidence.** The commit hashes anchor to that repository: all reader outputs, all judge verdicts and control receipts for every run in the ladder, the judge harness (frozen rubric, SHA-pinned templates, fail-closed transport), the tie-break rule and per-vote records, attribution tables, the gold-defect dossier, and run scripts. Re-scoring any run under the official judge costs ≈ $1.28 and requires only an OpenAI API key. The released/held boundary is frozen in `RELEASE_MANIFEST.md` (one SHA-256 per released file, regenerated deterministically by `scripts/build_release_manifest.py`): released are all reader checkpoints, judge verdicts, control receipts, agreement matrices, the dossier, the judge harness with its tests and rubric source, and the reader-lane scripts; held are the retrieval, rerank, packet-compiler and scaffold-operator sources and the materialized packets (benchmark haystack text). The held stages ship as pinned artifacts with verification receipts (determinism, gold-permutation, question-ID-sabotage, budget gates); every judge-side claim is re-derivable from the released checkpoints and harness alone. The benchmark data is MIT-licensed (`xiaowu0162/longmemeval-cleaned`), so the released checkpoints may carry question and gold text. Judge-side reproduction, run before release: the frozen pass-1 reader checkpoint was re-judged the same day in a clean output directory with the identical harness (`gpt-4o-2024-08-06`, resolved snapshot exact, fresh controls 12/12 and 12/12). Result **478/500**: 497 of 500 verdicts agree with the quoted 479; answerable is identical (452/470); the three flips are on byte-identical answer text (`7024f17c` and `031748ae_abs` lost, `a2f3aa27` gained). The official judge's own flip floor (§4.2) is therefore measured within a single judge on the headline pass, and "one pass above" is inside that floor; the tie is the claim. Receipts: `full-v34-opus_pass1/judge_gpt4o_repro_20260901/`.

9 Conclusion

A deterministic, negatively-controlled retrieval chain with the LLM confined to a replaceable reader measures 479/475 per 500 on LongMemEval-S — one pass above the published state of the art, the pair a statistical tie inside the instrument's noise band — under a measurement discipline that makes the number defensible at margins where measurement is usually the weakest link. The apparatus generalizes: freeze the promotion rule before the deciding run, control every improvement against what it might break, measure the judge, and publish the failures with the successes.

References

- Wu, D., et al. *LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory.* ICLR 2025, arXiv:2410.10813.
- Sen, S., Lumer, E., Gulati, A., Subbiah, V. K. *Chronos: Temporal-Aware Conversational Agents with Structured Event Retrieval for Long-Term Memory.* arXiv:2603.16862, Mar 2026.
- Barnes, T. *Observational Memory: 95% on LongMemEval.* Mastra Research, Feb 2026 (accessed 2026-08-31).
- Packer, C., et al. *MemGPT: Towards LLMs as Operating Systems.* 2023. Maharana, A., et al. *LoCoMo.* 2024.
- Nogueira, R., Cho, K. *Passage Re-ranking with BERT.* 2019. Chen, J., et al. *BGE M3-Embedding.* 2024. Cormack, G., et al. *Reciprocal Rank Fusion.* SIGIR 2009.
- Wang, X., et al. *Self-Consistency Improves Chain of Thought Reasoning.* ICLR 2023. Zheng, L., et al. *Judging LLM-as-a-Judge.* NeurIPS 2023. Dodge, J., et al. *Show Your Work.* EMNLP 2019.

Correspondence: Christopher J. Chanhnourack, Centennial Defense Systems. Evidence commits (in the repository named at publication): 477726726 · 04eb64763 · ad71321ea · ba805e575 · 08e5784bb · ad43cc99a · 9b0711c39.